

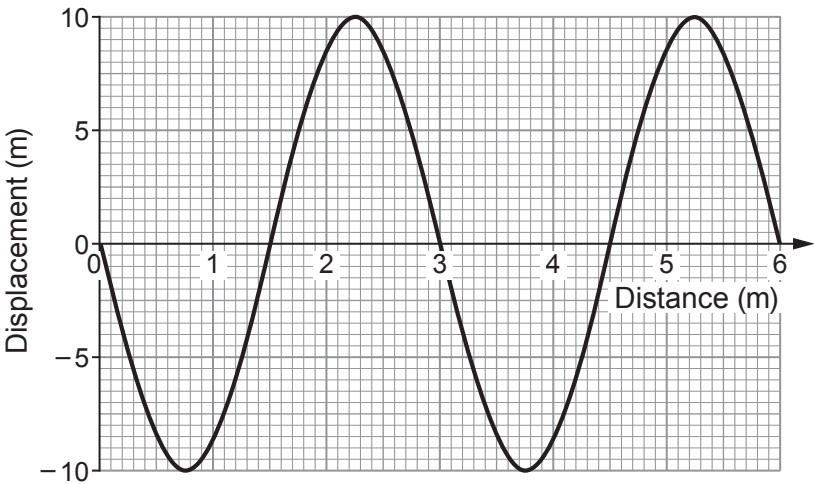
2. The diagram shows regions of the electromagnetic (em) spectrum.
The regions are in order.

radio waves	microwaves	infra-red	visible light	ultraviolet	X-rays	gamma rays
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(a) **Tick (✓) the *three* correct statements.** [3]

- The frequency of em waves decreases from radio waves to gamma rays.☐
- All em waves travel at the same speed in a vacuum.☐
- Gamma rays can be emitted from radioactive materials.☐
- The wavelength of em waves decreases from radio waves to gamma rays.☐
- Radio waves, microwaves, infra-red and visible light are all ionising radiations.☐
- All em waves are longitudinal.☐

(b) The graph below shows a wave diagram.



Use the following values to answer the questions below. You may use each value once, more than once or not at all.

2	3	5	6	10
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- (i) I. State the amplitude of the wave. m [1]
- II. State the number of complete waves shown. waves [1]
- III. State the wavelength of the wave. m [1]



(ii) I. Complete the following sentences by underlining the correct phrase in each bracket. [2]

The wave has a frequency of (7.5Hz / 7.5cm/s / 7.5cm).

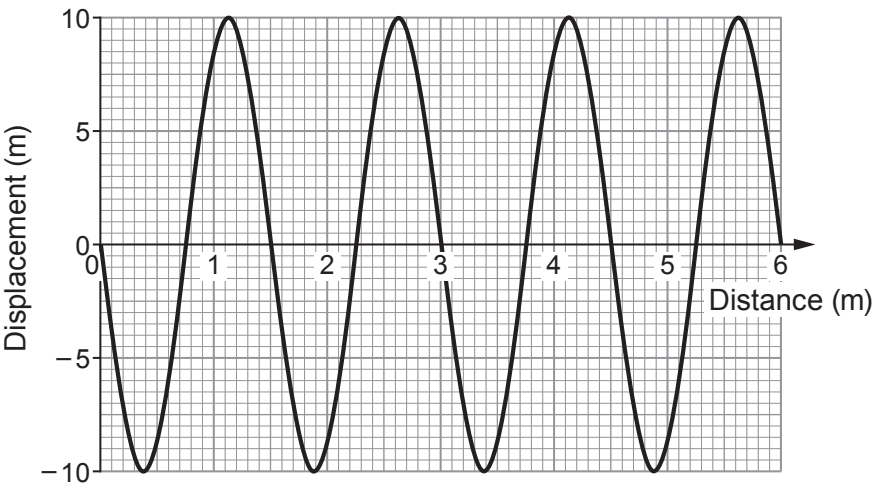
This means 7.5 waves pass a point each (second / minute / hour).

II. Use information from parts (b)(i) and (ii) to calculate the wave speed.
Use the equation:

wave speed = wavelength $\times$ frequency [2]

Wave speed = m/s

(c) The speed of the wave remains constant but its **frequency is doubled**.
A student draws a diagram of the expected wave. It is shown below.



Explain whether you agree with the student's diagram of the wave. [2]

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